Field Hours

CSCE 664 Wireless and Mobile Systems

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

Power, distance, and noise

You budget a wireless link before you invent a protocol. Transmit power spreads over space, and in free space received power falls with the square of distance. Double the distance, from 10 meters to 20, and received power falls to about 1/4. Noise does not fall with it, so the signal-to-noise ratio falls too. A frame that needed a higher ratio does not get through by wishing.

You also lose power to walls and to other senders in the same 20 MHz channel. A ratio you can compute is the threshold: if the radio needs a ratio of 10 to hold its modulation, and one doubling cut the ratio from 40 to 10, you are on the edge. One more doubling cuts it to 2.5, below 10. You then lower the bit rate or shorten the link. The MAC cannot repair a frame the receiver never heard.

Each doubling of distance cuts free-space power to a quarter. The noise floor stays.

Start at ratio 40

One doubling: ratio 10

Two doublings: ratio 2.5

Modulation needs ratio 10

Two doublings from a ratio of 40

1. Free-space power scales with one over distance squared. From 10 meters to 20, power is multiplied by 1/4, so a ratio of 40 becomes 10.

2. From 20 meters to 40, you multiply by 1/4 again. The ratio becomes 2.5.

3. A modulation that needs a ratio of 10 still works at 20 meters and fails at 40. You drop the bit rate or you do not use that link.

Problem 1. Distance doubles once in free space and the ratio starts at 40. What ratio do you expect?

Answer. 10. Received power falls to 1/4, and the noise did not fall, so 40/4 = 10.

Problem 2. The radio needs a ratio of 10 and you are at 2.5 after two doublings. What can the MAC do about a frame the receiver never hears?

Answer. Nothing useful. The frame never arrived, so a retry at the same rate still fails. You lower the bit rate or shorten the link until the ratio is at least 10.

Lecture 2 · 90 minutes

Carrier sense and a hidden sender

You share one channel by sensing energy before you transmit. A station that hears energy waits. The mechanism fails for a hidden terminal: station A and station C both reach the access point and do not reach each other. Each senses an idle medium and transmits. The access point sees the sum of 2 frames, and both die. Sensing at the sender did not sense the collision at the receiver.

You reserve the medium with a short request and a clear-to-send from the receiver. The request names a duration. Neighbors of the receiver hear the clear and set a network allocation vector for that many microseconds, here 200, and they stay quiet. Station C, hidden from A, hears the access point and defers. Without that clear, the data frame of 1500 bytes is what collides, which wastes more airtime than the short handshake of 2 control frames.

Sense protects a sender that can hear the other sender. It does not protect the access point.

Carrier sense

Listen, then send

Hidden pair both hear idle

Both frames die at the point

Request and clear

Receiver sends a clear

Neighbors defer 200 microseconds

Data of 1500 bytes is reserved

A and C both send

1. A and C cannot hear each other. Both hear the access point, sense idle, and transmit a 1500-byte frame.

2. The access point receives the sum of 2 frames. Both fail.

3. A clear from the access point, naming 200 microseconds, makes C defer even though C never heard A. The data frame is no longer the thing that collides.

Problem 1. A and C are hidden from each other and both sense idle. Where do the 2 frames collide?

Answer. At the access point, which hears both. Each sender's carrier sense was quiet because the other sender was out of range. The collision is at the receiver.

Problem 2. The clear tells neighbors to stay quiet for 200 microseconds. What does station C do if it hears that clear and not station A?

Answer. C defers for those 200 microseconds. The clear is how a hidden sender learns about a reservation it could not sense.

Lecture 3 · 90 minutes

Backoff and a growing window

You do not transmit the instant the medium goes idle. After a fixed interframe space you draw a backoff, a random count of slots from 0 through the contention window. A window of 15 means 16 possible counts, 0 through 15. You decrement the count only while the medium is idle. When it hits 0 you transmit. Two stations that draw the same count collide, and both see a failure.

You double the window after a collision so the next draw spreads wider. A window of 15 becomes 31, then 63, up to a cap of 1023. A success shrinks it, often back to 15. If 20 stations all start at 15, many draw the same slot, and skipping the doubling keeps that clash. If you start at 1023, you pay a long wait even when you are the only sender.

The count falls only on idle slots. A collision widens the next draw.

1. Wait the interframe gap

2. Draw from 0 to 15

3. Send when the count hits 0

4. Double toward 1023

A collision, then a wider draw

1. Two stations draw the same count inside a window of 15. There are 16 possible counts, 0 through 15, and they picked one.

2. They transmit together and both fail. Each sets the window to 31.

3. The next draws are spread over 32 counts, 0 through 31. A later success can shrink a window back toward 15. Starting at 1023 would have delayed even a single sender.

Problem 1. The contention window is 15. How many distinct backoff counts can you draw?

Answer. 16. The counts are 0 through 15. You send only when the count reaches 0, and you decrement only while the medium is idle.

Problem 2. After one collision the window was 15. What is it for the next draw, and what is the cap?

Answer. 31 for the next draw. Each further collision doubles it, and the cap in this lecture is 1023. A success can bring it back toward 15.

Lecture 4 · 90 minutes

Loss that is not a full queue

You inherited a transport rule that treats loss as congestion. A timeout, or 3 duplicate acknowledgements, cuts the congestion window in half. A window of 64 segments becomes 32. On a wire, that reaction matches a full queue. On a radio, a bit error or a hidden-terminal collision drops a frame the queue did not cause. You still cut the window, so a loss rate of 2 percent can pin the rate far below an idle medium.

You repair the mistake in 1 of 2 places. Link retransmission hides the loss: the access point retries, and the

window stays at 64 if the retry beats the timeout. The retry must keep order, or 3 duplicate acknowledgements still cut the window to 32. The other repair is a transport that does not halve when the link marks the loss as wireless. You name which one you used.

The transport sees a gap. It does not see whether the queue or the air caused it.

Wired assumption

Loss means a full queue

Window 64 becomes 32

The cut is the right reaction

Radio loss

A collision, not a queue

Same cut, wrong reason

Retry at the link, or mark it

A hidden collision cuts the window

1. The congestion window is 64 segments. A hidden-terminal collision drops one frame. The link does not retry.

2. The sender sees loss, by timeout or by 3 duplicate acknowledgements, and sets the window to 32.

3. No queue overflow was required. A link retry that finishes in time would have left the window at 64.

Problem 1. The window is 64 and a radio loss is visible to the transport. What does the window become?

Answer. 32. Loss halves it. On a wire that matches a full queue. On a radio the frame may have died in a collision, and the cut still happens unless something hides the loss.

Problem 2. The access point retries the frame and the retry finishes before the transport timeout. What happens to a window of 64?

Answer. It stays 64, because the transport never saw the loss. If the retry reorders segments badly enough to cause 3 duplicate acknowledgements, the window still falls to 32.

Lecture 5 · 90 minutes

Bits per hertz, then users

You estimate cell capacity as bandwidth times bits per hertz, then divide by the users who share the air. A 20 MHz channel at 4 bits per hertz carries 80 million bits per second before headers. Ten active users, sharing fairly and never colliding, get 8 million bits per second each. Backoff slots and interframe spaces eat into the 80. You quote the 8 only with those taxes named.

You do not use 4 bits per hertz when the ratio cannot hold it. Two doublings of distance cut power by 1/16. A starting ratio of 40 becomes 2.5, and a modulation that needs 10 will not run. You drop that user to 1 bit per

hertz. A fair tenth of a 20 MHz channel is then 2 million bits per second, not the 8 million the cell center can take.

Raw rate first, then the share, then the ratio the edge user actually has.

20 MHz times 4 -> 80 million bits -> Divide by 10 users -> Cut if the ratio fails

Center user and edge user

1. The channel is 20 MHz at 4 bits per hertz, so the raw rate is 80 million bits per second.

2. Ten fair users with no collisions each get 8 million, before backoff and headers take their cut.

3. An edge user at ratio 2.5 cannot hold 4 bits per hertz. At 1 bit per hertz their fair tenth is 2 million bits per second, because 20 million divided by 10 is 2 million.

Problem 1. A 20 MHz channel runs at 4 bits per hertz and 10 users share it fairly, ignoring headers. What does each user get?

Answer. 8 million bits per second. The raw rate is 20 times 4 = 80 million, and 80/10 = 8 million. Backoff and interframe gaps reduce that 8 further.

Problem 2. The same user can hold only 1 bit per hertz and still receives a tenth of the airtime. What is that share?

Answer. 2 million bits per second. The whole channel at 1 bit per hertz is 20 million, and a tenth is 2 million. They do not get the 8 million that 4 bits per hertz would allow.

Lecture 6 · 90 minutes

Hops and a duty cycle

You route in an ad hoc net with no access point. A source floods a route request. Each forwarder records the previous hop and adds 1 to the hop count. The destination replies along the reverse path. A data route of 4 hops costs 4 transmissions of the payload, and a broken link costs a new flood. You flood the short request once. You do not flood the data.

You treat energy as the budget on a sensor. A node awake 10 milliseconds each second has a duty cycle of 1 percent, so a 100 milliwatt radio averages 1 milliwatt if sleep draws nothing. A route of 4 hops multiplies one report by 4, because each hop transmits. You pick a parent toward the sink and sleep the rest of the second. A beacon every 10 milliseconds at full power erases the duty cycle you computed.

Flood the request, not the data. Then sleep. A chatty beacon spends the battery.

1. Flood a short request

2. Reply along reverse hops

3. Unicast a 4-hop payload

4. Sleep the other 990 ms

Four hops and a 1 percent radio

1. The route request collects a path of 4 hops. Each forwarder added 1 to the hop count.

2. The payload is unicast 4 times, once per hop, not flooded.

3. The radio is on for 10 milliseconds out of 1000, a duty cycle of 1 percent. At 100 milliwatts while awake and 0 while asleep, the average is 1 milliwatt. Four hops make one report cost about 4 transmissions.

Problem 1. A route has 4 hops. How many times is the payload transmitted, and what do you flood instead?

Answer. The payload is transmitted 4 times, once per hop. You flood the short route request once, not the data. A break costs a new flood of that request.

Problem 2. The radio draws 100 milliwatts and is awake 10 milliseconds each second. What is the duty cycle, and what is the average draw if sleep draws 0?

Answer. The duty cycle is 1 percent, because 10/1000 = 0.01. The average draw is 1 milliwatt. A beacon every 10 milliseconds at full power removes that saving.

Lecture 7 · 90 minutes

Management frames and a second radio

You protect data frames and you often leave management frames open. An attacker sends one deauthentication, spoofing the access point, and the client leaves a network it was using. No data key is required for that management frame on older gear. The client then scans and may join an evil twin that answers the same network name. You lose the session at frame 1 of the attack, before any guess at the password.

You authenticate the infrastructure, not only a password. The handshake binds the key to both names and checks an integrity code on management frames, so 1 forged deauthentication fails. If you turn that check off for a lab of 15 laptops, any laptop can forge the frame. Data protection does not save the client: the attacker aims them at another radio. You check the access point name and the management code together.

Encrypting the payload does not authenticate the access point that sent a management frame.

Open management

One forged deauthentication

Client leaves at frame 1

Evil twin reuses the name

Checked management

Integrity code on the frame

Forgery fails the check

Lab of 15 must keep it on

One deauthentication

1. The client is associated. The attacker sends 1 deauthentication that names the access point. Older gear does not check a data key on that frame.

2. The client drops the association and scans. An evil twin answers the same network name.

3. With an integrity check on management frames, that 1 frame fails the check and the client stays. Turning the check off for 15 lab machines restores the attack.

Problem 1. Data frames are encrypted and management frames are not checked. How many forged deauthentication frames does it take to drop the client?

Answer. 1. The client leaves on that frame. Encryption of the data does not apply to the management frame, so no password guess is required.

Problem 2. You disable the management integrity check for a lab of 15 laptops. What can any one of those laptops do?

Answer. Forge a deauthentication and aim a client at another radio. The check is what makes that frame fail. Data-frame protection does not replace it.

Lecture 8 · 90 minutes

One frame, four layers

You can now follow one frame from the air to the transport and name the field that failed. You can compute a drop to 1/4 of the power when distance doubles, explain a collision that carrier sense missed, and double a contention window from 15 to 31. You can say why a loss cut a window from 64 to 32, and why a sensor awake 10 milliseconds per second sits at a 1 percent duty cycle.

You can also put a number on the cell and on the attack. A 20 MHz channel at 4 bits per hertz is 80 million bits per second before sharing, and 10 users do not each receive all of it. One forged deauthentication moves a client if management frames are unchecked. You can write which layer you would change, and you can predict the side effect one layer up, including a window cut that a queue did not cause.

Name the layer that failed before you retune a different layer.

Ratio after the distance

Backoff and the hidden pair

Window cut on loss

Management integrity

Collision, then a halved window

1. A and C are hidden, both sense idle, and their frames collide at the access point. The link does not retry.

2. The transport window is 64. It sees the loss and becomes 32. No queue had to overflow.

3. A clear that reserves the medium, or a link retry that beats the timeout, would have kept the window at 64. The radio failure and the transport reaction are one story.

Problem 1. A hidden terminal causes a collision, the link does not retry, and the congestion window is 64. What does the window become, and did a queue overflow?

Answer. The window becomes 32, because a visible loss halves it. No queue overflow is required. Both senders sensed an idle medium, and the frames collided at the access point.

Problem 2. Distance has doubled twice, so free-space power is 1/16 of the start. The starting ratio was 40. The channel is 20 MHz, the edge user holds only 1 bit per hertz, and they get a fair tenth of the airtime. What ratio do they see, and what share do they get?

Answer. The ratio is 40/16 = 2.5. At 1 bit per hertz the full channel is 20 million bits per second, and a tenth is 2 million. They do not get the 8 million that 4 bits per hertz would give a center user.